

MODEL VERSION SELECT

Clean Dataset - Train Set : S1 (5000), S2 (50000)

Real Dataset - Test Set : S3 (55000), S4 (105000)

Pair	Version	Features	Clean Accuracy	Real Accuracy	Gap	Real Predict Runtime	Pipeline Runtime
S1 5,000 vs S3 55,000	V1	25	80.60%	80.09%	-0.51 pp	1.5294s	0.6297s
S1 5,000 vs S3 55,000	V2	64	81.20%	80.71%	-0.49 pp	0.8747s	30.5754s
S1 5,000 vs S3 55,000	V3	81	81.90%	82.04%	+0.14 pp	0.6700s	33.0476s
S1 5,000 vs S3 55,000	V4	107	82.50%	82.05%	-0.45 pp	1.0841s	33.4670s
S1 5,000 vs S3 55,000	V5	64	82.00%	81.91%	-0.09 pp	0.6660s	33.4400s
S2 50,000 vs S4 105,000	V1	25	82.29%	82.45%	+0.16 pp	0.7325s	2.1396s
S2 50,000 vs S4 105,000	V2	64	82.17%	81.92%	-0.25 pp	1.0347s	230.2552s
S2 50,000 vs S4 105,000	V3	81	83.04%	82.87%	-0.17 pp	1.0500s	254.8428s
S2 50,000 vs S4 105,000	V4	107	83.16%	82.86%	-0.30 pp	1.2532s	258.5415s
S2 50,000 vs S4 105,000	V5	64	82.86%	82.78%	-0.08 pp	0.9904s	258.4398s

Clean Dataset S1 vs S2

Version	Features	S1 Accuracy	S2 Accuracy	Gap	S1 Predict	S2 Predict	S1 Pipeline	S2 Pipeline
V1	25	80.60%	82.29%	+1.69 pp	0.1390s	0.3488s	0.6297s	2.1396s
V2	64	81.20%	82.17%	+0.97 pp	0.0795s	0.4927s	30.5754s	230.2552s
V3	81	81.90%	83.04%	+1.14 pp	0.0609s	0.5000s	33.0476s	254.8428s
V4	107	82.50%	83.16%	+0.66 pp	0.0986s	0.5968s	33.4670s	258.5415s
V5	64	82.00%	82.86%	+0.86 pp	0.0605s	0.4716s	33.4400s	258.4398s

Real Dataset S3 vs S4

Version	Features	S3 Accuracy	S4 Accuracy	Gap	S3 Predict	S4 Predict	S3 Pipeline	S4 Pipeline
V1	25	80.09%	82.45%	+2.36 pp	1.5294s	0.7325s	2.3536s	4.4932s
V2	64	80.71%	81.92%	+1.21 pp	0.8747s	1.0347s	253.2807s	483.5358s
V3	81	82.04%	82.87%	+0.83 pp	0.6700s	1.0500s	280.3271s	535.1699s
V4	107	82.05%	82.86%	+0.81 pp	1.0841s	1.2532s	284.3956s	542.9370s
V5	64	81.91%	82.78%	+0.87 pp	0.6660s	0.9904s	284.2838s	542.7235s

สาเหตุที่ Run-Time บาง Version ใช้เวลานานกว่าเป็นเพราะ :

1. การ Join หลาย Table เช่น table customer, table product
2. คำนวณ Customer History (ประวัติลูกค้า) รายการ/order เมื่อลูกค้ามีหลายรายการ
3. คำนวณ ประวัติลูกค้าย้อนหลัง เช่น 7/30/90/365 วัน
4. คำนวณ return rate ราย product/category/courier/payment
5. จำนวน Feature ที่เพิ่มมาทำให้ model ทำงานหนักขึ้น

Version	Features	Clean S1 5,000	Real S3 55,000	Gap S3-S1	Clean S2 50,000	Real S4 105,000	Gap S4-S2
V1	25	80.60%	80.09%	-0.51 pp	82.29%	82.45%	+0.16 pp
V2	64	81.20%	80.71%	-0.49 pp	82.17%	81.92%	-0.25 pp
V3	81	81.90%	82.04%	+0.14 pp	83.04%	82.87%	-0.17 pp
V4	107	82.50%	82.05%	-0.45 pp	83.16%	82.86%	-0.30 pp
V5	64	82.00%	81.91%	-0.09 pp	82.86%	82.78%	-0.08 pp